


CVG

Company Confidential

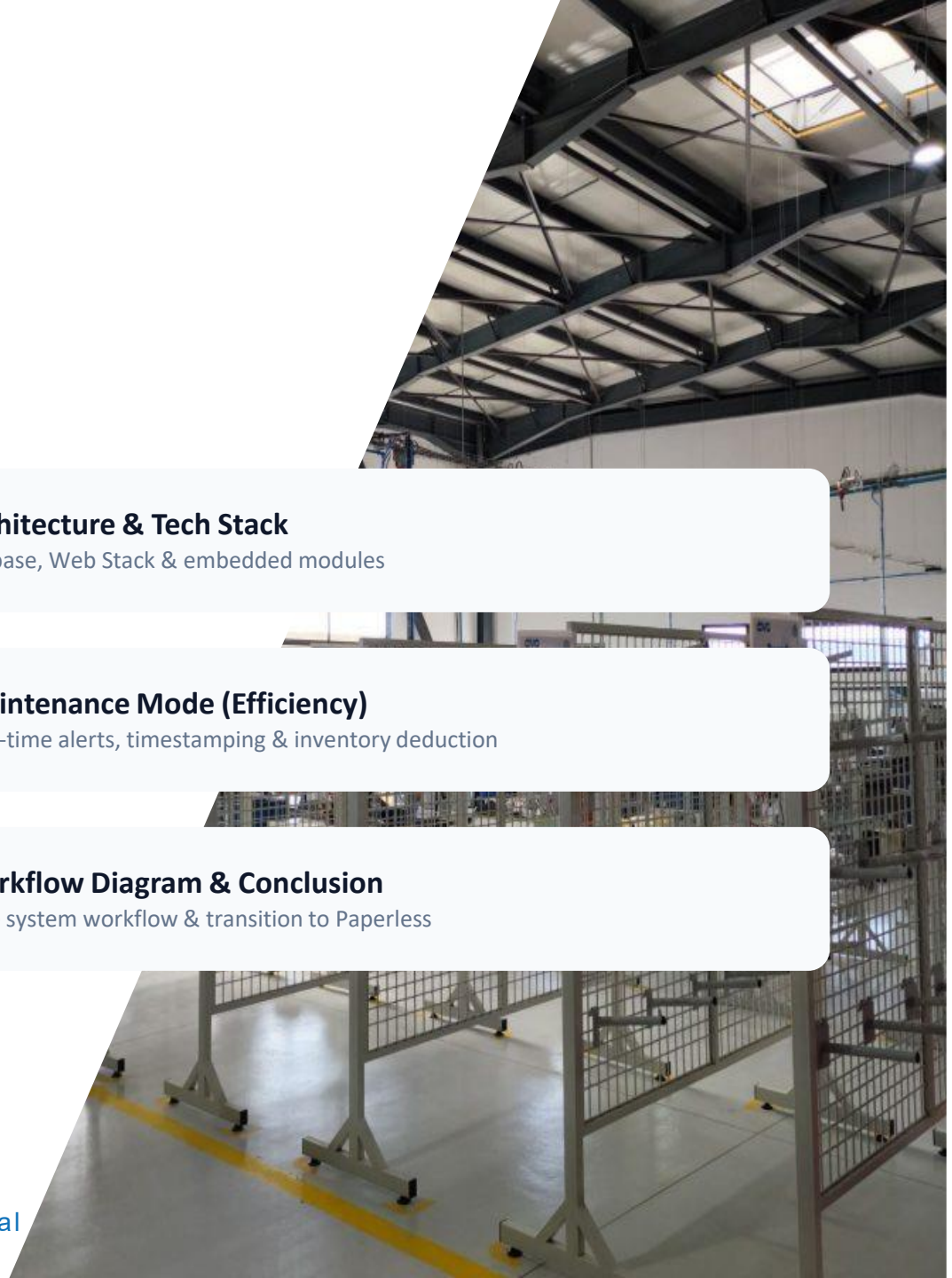
# Integrated Maintenance Digitization Platform

***By : Mustapha Lamriss***

Ethics | Execution | Enterprise | Excellence | Engagement

PRESENTATION PLAN

# Executive Summary



**01** **Main Menu**
Global vision & workflow centralization

**02** **Architecture & Tech Stack**
Firebase, Web Stack & embedded modules

**03** **Operator Mode (Reactivity)**
QR Code scanning & instant breakdown alerts

**04** **Maintenance Mode (Efficiency)**
Real-time alerts, timestamping & inventory deduction

**05** **Manager Mode (Analytics & Control)**
KPIs (MTTR/MTBF), charts & stock management

**06** **Workflow Diagram & Conclusion**
Core system workflow & transition to Paperless

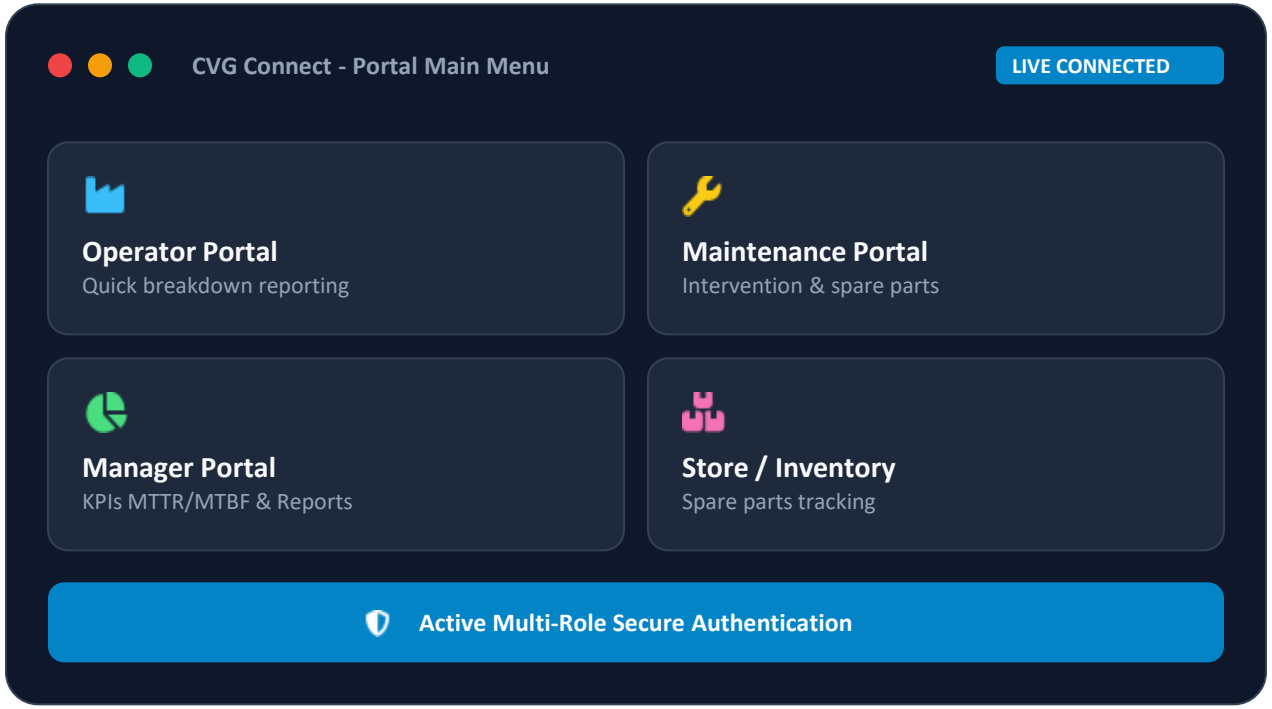
# Integrated Maintenance Digitalization Platform

## Workflow Centralization

Direct, seamless connection between Production, Maintenance, and Management via a single instant interface.

## Industrial Performance

Global operational efficiency optimization and drastic reduction of unplanned machine downtime.



# Ecosystem & Real-Time Web Application

### Database

**Firebase Real-time DB**

Instant data synchronization across all users without page refreshes.

### Modern Web Stack

**HTML5, CSS3, JS**

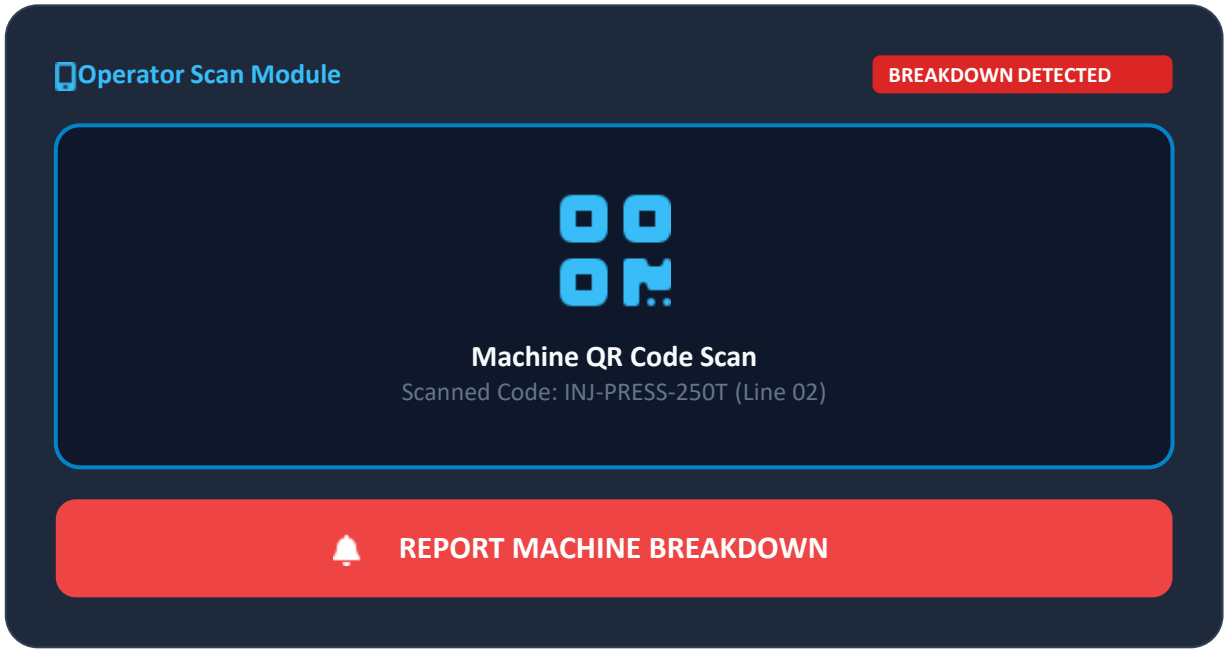
Responsive, lightweight UI adapted for both desktop stations and mobile devices.

### Embedded Tools

**Scanner & Reporting**

Built-in QR Code/barcode scanner and automated Excel report export engine.

# Instant Scan & Breakdown Reporting




### Scan & Identification

Instant machine and production line identification with zero manual entry errors.

### Line Stop Alert

One-click breakdown report triggering instant audio and visual alerts for technicians.

# Response Handling, Timestamping & Spare Parts

### 1. Alert Dashboard

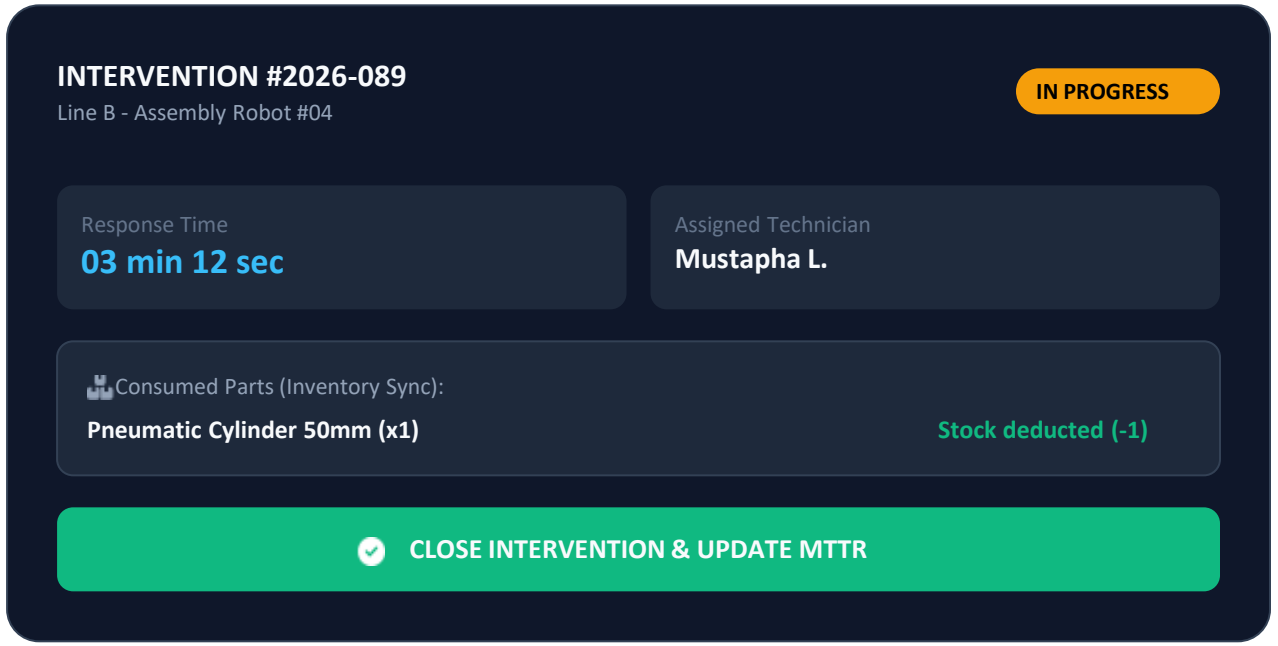
Visual and audible tracking of pending shopfloor breakdowns.

### 2. Timestamped Response

Automatic timestamping upon acknowledgement (RECEIVED) & reaction time calculation.

### 3. Closure & Stock Deduction

Direct inventory link upon ticket closure (automatic deduction of parts used).



# Key Performance Indicators (KPIs), Analytics & Inventory Control




**MTTR (Mean Time To Repair)**
Measurement of technical intervention efficiency.

**MTBF (Mean Time Between Failures)**
Measurement of overall machine reliability.

**Inventory Control**
Automated alerts for critical low-stock thresholds.

# Workflow Diagram (The Core System)

| | |
|---|---|
| EVENT | **Machine breakdown detected by Operator**: QR Code scan & instant report via CVG-Connect app. |
| ACTION | **Firebase logging & Alert**: Instant visual/audible notification sent to technician on call. |
| REACTION | **Acknowledgment & Intervention**: Click "RECEIVED" (timestamped), field repair & parts entry. |
| RESULT | **Closure & Cloud Sync**: Instant recalculation of MTTR/MTBF, auto stock deduction & Excel export. |

# Traditional Paper Issues vs Digitalized Solution

## Legacy Method (Paper)

**Information Loss:** Paper stoppage forms frequently misplaced or damaged.

**Impossible Analysis:** Inability to extract reliable KPIs from manual logbooks.

**Slow Reaction:** Significant delay between paper logging and actual maintenance response.

## CVG-Connect (Paperless)

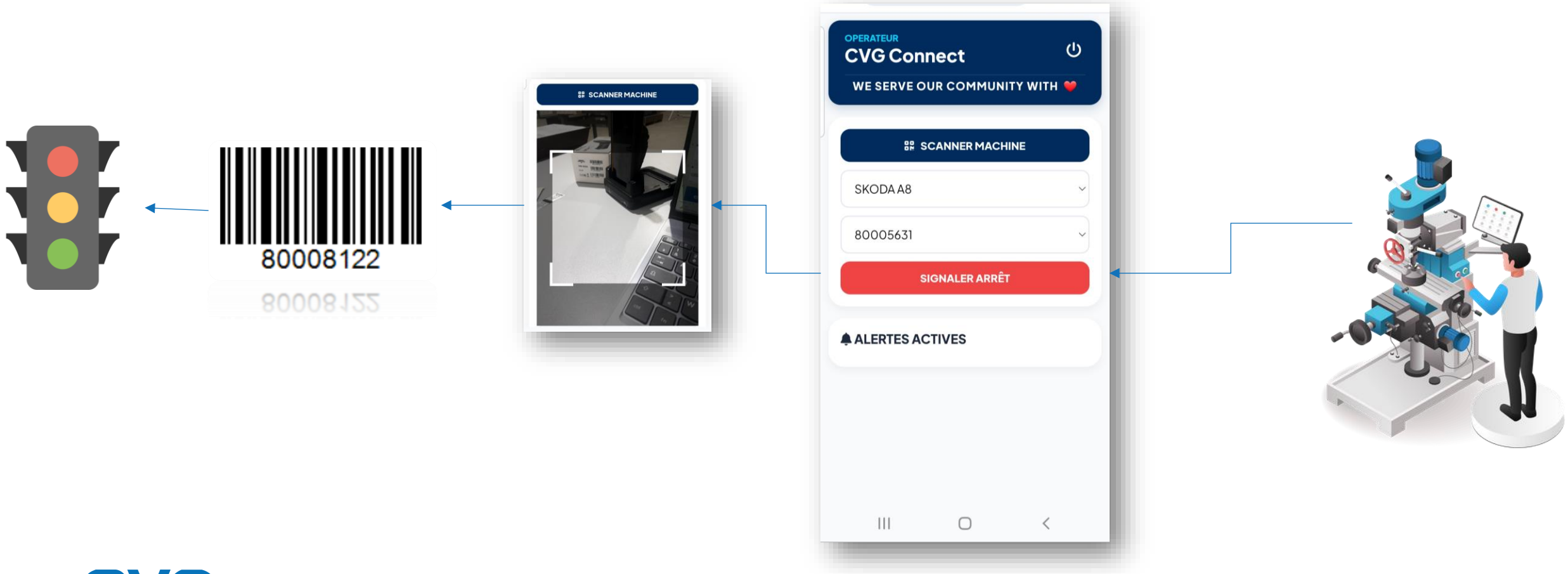
**Digital Alerting:** Immediate QR Code reporting on workstation or mobile.

**Cloud Archiving:** Secure automatic backups on Firebase cloud storage.

**Full Traceability:** Precision second-by-second timestamping & dynamic MTTR/MTBF calculation.

# OPERATOR mode (Responsiveness)

- **Scan & Signal :** Instant machine identification via QR code.
- **Shutdown Alert:** One-click failure reporting with immediate notification.

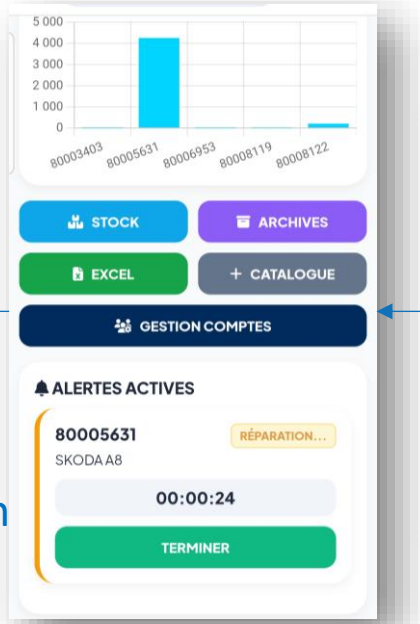


# MAINTENANCE Mode (Efficiency)

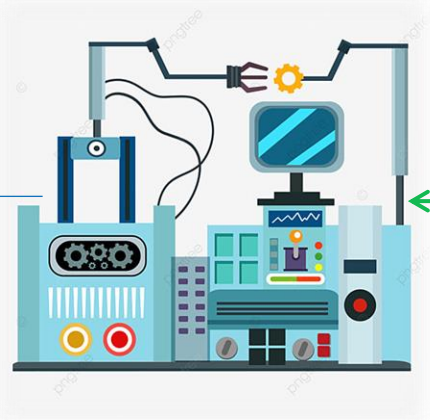
- **Alert dashboard:** Visual and audible monitoring of pending outages
- **Support:** Automatic timestamp of the reception and closure of the intervention.
- **Document management:** Direct link to inventory at closing (automatic deduction of quantities).



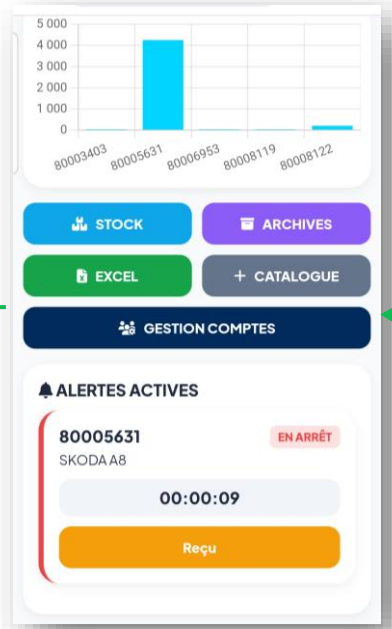

Closure of intervention




Finish the intervention



Machine OK




Received by the technician



Machine Shutdown

Treatment & Synchronization Data



Scan QR Code

Maintenance

Cloud

Management





- **Alert :** Receive a visual and audible notification on the dashboard
- **Pickup:** The technician clicks on "RECEIVED"
  (Timestamp of the start of the intervention)
- **Intervention :** Field diagnostics and repair

- **Stock :** Automatic deduction of spare parts.
- **KPIs :** Instant MTTR and MTBF recalculation.
- **Rapport :** Automatic generation of an Excel for reporting

- **Action :** The technician closes the intervention with the number, the solution, and the part used.




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# END

